

NIHESH RACHAKONDA

AI Engineer

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Portfolio | GitHub | LinkedIn | LeetCode

AI engineer and software engineering intern who ships production generative-AI systems end to end: LLM orchestration, RAG over vector and knowledge-graph stores, and intelligent agents with tool-using workflows, built on Python and FastAPI services across AWS and Azure with Docker and CI/CD. National hackathon winner with strong backend, API, and distributed-systems fundamentals, and a habit of analyzing logs, metrics, and test results to find issues and optimize performance.

EDUCATION

VNR Vignana Jyothi Institute of Technology	Hyderabad, Telangana
B.Tech in Information Technology	2023 – 2027 CGPA: 9.16 / 10
Sri Chaitanya Junior College	Hyderabad Intermediate (MPC) 2021 – 2023 96.7%

EXPERIENCE

Dattam Labs Software Engineer Intern	Hyderabad Jan 2025 – Present
- Engineered high-throughput data pipelines for AeroLeads using AWS Lambda, SQS message queuing, and Elasticsearch indexing to power real-time lead intelligence at scale. - Built AccessLife, a cross-platform mobile app (Expo) with a FastAPI backend on Azure VMs serving multiple US school districts; owned the CI/CD pipelines that automate deployments across Azure and AWS.	

SKILLS

Languages:	Python, SQL, Java, JavaScript / TypeScript, C++, Go, Dart
GenAI & ML:	LLMs (GPT-4o, Gemini, LLaMA3), RAG pipelines, Intelligent agents (Google ADK / A2A, MCP), LangChain, Embeddings, TensorFlow
Cloud & DevOps:	AWS (Lambda, SQS, S3, EC2, ECR, Bedrock), Azure, Docker, CI/CD (GitHub Actions), Nginx, Linux
Backend & APIs:	FastAPI, Express, NestJS, REST API design, Pydantic, SQLAlchemy, JWT / OAuth, WebSockets, microservices & distributed systems
Databases & Vector:	PostgreSQL, pgvector (vector DB), Neo4j (knowledge graph), MongoDB, Redis, Elasticsearch
Tools & Methods:	Git, pytest / Vitest, load & latency testing, Agile

PROJECTS

GST Reconciliation Engine (Knowledge-Graph RAG)	GitHub
<i>Python, FastAPI, Neo4j, OpenAI / Gemini, Docker</i> Built a tax-reconciliation engine exposing 17+ REST APIs that models GST returns as a Neo4j knowledge graph to flag mismatches and circular-trading fraud, with a RAG assistant and LLM-generated audit trails over an OpenAI / Gemini / Ollama fallback chain, deployed with Docker behind Nginx. Won INR 50,000 at the National ACM-KLH HackWithAI 2026.	
PDF Summarizer + RAG Chat (Vector DB Pipeline)	GitHub
<i>FastAPI, Postgres + pgvector, OpenAI, AWS S3</i> Built a RAG pipeline that chunks and embeds PDFs (OpenAI text-embedding-3-small, 1536-dim) into a pgvector vector database and serves grounded gpt-4o-mini answers with page-range citations via cosine-similarity search across 10 JWT-secured REST endpoints, with object storage on AWS S3.	
Phoenix AI Toolkit (Multi-LLM GenAI Platform)	Live
<i>Node.js, Express, MongoDB, LangChain, Gemini</i> Built a generative-AI platform that orchestrates Gemini 2.5 Flash, Imagen 3, and Ollama LLaMA3 via LangChain to auto-generate sites, emails, and images alongside a context-aware chatbot, using vector similarity search for feedback insights behind a secured REST API with JWT auth and rate limiting; deployed live.	

ACHIEVEMENTS & CERTIFICATIONS

- Winner, GDG TechSprint 2026: **Civic Lemma**, an AI civic-grievance platform (MobileNetV2 classifier + Gemini)
- Winner, AI Week Vibe Coding 2026: **SEFS**, a smart file organizer with a RAG chatbot
- AWS Open Data contributor: Indian Court Judgements scraped into AWS S3 | GitHub
- Research Paper: Model Selection via Hypothesis Testing | IJFMR
- Database Structures & Management with MySQL | Meta (Coursera) | Hacktoberfest 2024 L4